

Naraen Palanikumar

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SUMMARY

Bioinformatics engineer with 3+ years of research and engineering experience building production-grade genomics infrastructure. Designs containerized Nextflow workflows, cloud-native architectures on GCP, and API layers that take sequencing data from ingestion to interpretation. Google Cloud Certified Professional Data Engineer.

TECHNICAL SKILLS

Languages: Python, R, SQL, Bash, MATLAB, JavaScript, TypeScript, HTML/CSS

Workflow & Cloud: Nextflow (DSL2), Docker, Git, containerization, CI, conda, pytest, reproducible pipelines, Google Cloud Platform, Cloud Run, Vertex AI, IAM, cloud-native architecture, data engineering

Genomics: genomics, transcriptomics, proteomics, metabolomics, 10x Genomics, variant calling, bulk RNA-seq, differential expression, GIAB/HG002 benchmarking, GATK4, HaploTypeCaller, joint genotyping, BWA-MEM, HISAT2, samtools, bcftools, featureCounts, fastp

Single-Cell & Spatial: Seurat, Scanpy, Cell Ranger, tcrdist3, UMAP, clustering, pseudotime, trajectory inference, anchor-based integration, reference-guided annotation, AnnData, Squidpy

Statistics & ML: meta-analysis, survival analysis, statistical validation, SciPy, pandas, NumPy, scikit-learn, negative-binomial modeling, hypergeometric enrichment, multiple-testing correction, PyTorch, machine learning, LLM/RAG architecture, dimensionality reduction

Wet Lab: spectral flow cytometry (Cytek Aurora), FlowJo, flow panel design, compensation, gating strategy, PCR, qPCR, mammalian cell culture, aseptic technique

EXPERIENCE

Founding Engineer

Mar 2026 – Present

Phalaena Automata

Austin, TX

- Spearheading end-to-end development of a biotechnology research accelerator platform, managing the product lifecycle from initial architecture to cloud deployment.
- Architected containerized Nextflow DSL2 workflows executing on Cloud Run under least-privilege IAM, with a FastAPI ingestion layer normalizing heterogeneous sequencing inputs.
- Validated variant-calling output against the GIAB HG002 benchmark to establish pipeline accuracy on a public ground-truth reference.
- Developing full-stack AI-centric infrastructure on Google Cloud, integrating RAG and LLM architecture to automate unstructured data interpretation pipelines and accelerate the sequence-to-insight loop.

Bioinformatics Engineer (Independent Projects)

Jul 2025 – Mar 2026

Independent Research & Development

Remote

- Developed cloud-native, containerized bioinformatics pipelines for high-throughput omics workflows, applying Google Cloud Professional Data Engineer practices to reproducible research infrastructure.
- Built end-to-end single-cell and bulk RNA-seq analyses on public oncology datasets, covering QC, integration, cell-type annotation, trajectory inference, and survival analysis.
- Published analyses as interactive dashboards and open project write-ups to make results explorable by non-computational audiences.

Research Assistant

Oct 2024 – Jun 2025

Malloy Lab, Uniformed Services University of the Health Sciences

Bethesda, MD

- Led bioinformatics analysis to extract and analyze CD4 T-cell repertoires from scTCR-seq data, contributing to published research in *Frontiers in Immunology*.
- Bioinformatically isolated and modeled CD4 T-cell populations using custom extraction scripts, comparing TCR repertoire usage across participants stratified by COVID-19 severity and vaccination status.
- Executed molecular and cell-based assays end to end, from sample receipt and processing through staining, acquisition, and downstream analysis, bridging bench work and computational output within a single workflow.

Undergraduate Research Assistant

Aug 2021 – Jun 2023

Sweeney Lab, University of Illinois Urbana-Champaign

Champaign, IL

- Built and launched an interactive R Shiny application to streamline fiber photometry data processing and interpretation, contributing to published research in *The Journal of Neuroscience*.

- Analyzed complex scRNA-seq datasets using Cell Ranger and Seurat to uncover metabolism-associated neural circuits and rare cell populations.
- Consolidated a multi-step manual analysis into a single reactive application, cutting photometry processing from a multi-hour scripting task per session to minutes and enabling non-programmer lab members to run their own analyses.

SELECTED PROJECTS

Germline Variant Calling (GATK short-variant pipeline) | Nextflow, GATK4, BWA-MEM, samtools, bcftools | writeup

- Twelve-stage Nextflow DSL2 germline short-variant pipeline following the GATK best-practice path, from BWA-MEM alignment and duplicate marking through HaplotypeCaller GVCF calling, joint genotyping, hard-filtering, and consequence annotation to a filtered cohort VCF.
- Built a synthetic diploid cohort with planted germline variants as the offline validation path, achieving perfect recovery of all 24 planted sites (recall = precision = 1.00, genotype concordance 1.00 across 144 sample-genotypes) via an automated self-check that gates every run.
- Wired a GIAB HG002 benchmarking path against the NIST v4.2.1 high-confidence truth set, range-slicing reads directly out of GIAB's chr20 BAM over HTTPS so only the bytes for the target window are fetched.

Spatial Transcriptomics Deconvolution (10x Visium breast cancer) | Nextflow, Python, scanpy, anndata, squidpy | writeup

- Nine-stage Nextflow DSL2 pipeline mapping cell types into tissue space from a 10x Visium breast-cancer section (3,798 spots) against a 100,064-cell annotated scRNA-seq reference, covering QC, shared-gene-space normalization, signature construction, per-spot deconvolution, and spatially variable genes.
- Implemented per-spot deconvolution as non-negative least squares with inverse-mean gene weighting to keep the fit from collapsing onto high-expression immune signatures under cross-platform reference shift, recovering the expected tumor composition with cancer epithelium leading in 2,201 of 3,798 spots.
- Computed spatially variable genes by a from-first-principles Moran's I on a symmetric kNN spot graph, vectorized across genes with a seeded permutation test, identifying 949 significant genes that resolve into epithelial, stromal, and immune compartments.

Plant Bulk RNA-seq (rice drought tolerance, Nextflow) | Nextflow, HISAT2, featureCounts, pydeseq2, Python | writeup

- Ten-stage Nextflow DSL2 bulk RNA-seq pipeline taking raw Illumina reads from drought-tolerant and drought-susceptible rice cultivars through trimming, spliced HISAT2 alignment (83-94% aligned across 38,993 quantified genes), featureCounts quantification, differential expression, and GO over-representation.
- Modeled the genotype contrast with pydeseq2 on a ~condition + genotype design so tolerance is estimated controlling for drought treatment, resolving 56 genes at $p_{adj} < 0.05$ that concentrate in defense-response and diterpenoid (phytoalexin) biology.

PUBLICATIONS

- Parsons E, Lu Z, Richard SA, Zelkoski A, Le J, Palanikumar N, Nguyen P, Alba C, Sukumar G, Rosenberger J, Zhang X, Burgess TH, Colombo R, Mende K, Berjohn C, Epsi N, Agan BK, Tribble D, Lindholm DA, Dalgard CL, Pollett SD, Malloy AMW, and EPICC COVID-19 Cohort Study Group. Patterns of restricted TCR usage following SARS-CoV-2 vaccination and severe disease. *Frontiers in Immunology*. 2025;16:1576903. doi:10.3389/fimmu.2025.1576903
- Cho D, O'Berry K, Possa-Paranhos IC, Butts J, Palanikumar N, Sweeney P. Paraventricular Thalamic MC3R Circuits Link Energy Homeostasis with Anxiety-Related Behavior. *The Journal of Neuroscience*. 2023 Sep 6;43(36):6280-6296. doi:10.1523/JNEUROSCI.0704-23.2023. PMID: 37591737. PMCID: PMC10490510.

EDUCATION

Master of Engineering (M.Eng.), Bioengineering

University of Illinois Urbana-Champaign

Aug 2023 – May 2024

Urbana-Champaign, IL

Bachelor of Science (B.S.), Bioengineering

University of Illinois Urbana-Champaign

Aug 2019 – May 2023

Urbana-Champaign, IL

CERTIFICATIONS

Google Cloud Certified, Professional Data Engineer (Dec 2025 to Dec 2027)